REMI

Document Chatbot - Project Overview

What is REMI?

REMI is a lightweight, intelligent document chatbot. Upload a PDF, Word document, PowerPoint, or text file and REMI instantly summarizes it and lets you ask follow-up questions in plain English.

What does it do?

- Ingests documents: PDF, DOCX, PPTX, TXT, and Markdown.
- Auto-generates a concise summary with key bullet points.
- Answers questions using only the content of your uploaded document.
- Handles vague follow-up questions by rewriting them into better search queries.
- Cites page and slide numbers so you can verify every answer.

How it works

Instead of sending the entire document to the language model on every question, REMI uses Retrieval-Augmented Generation (RAG). The document is split into overlapping chunks, embedded with a local sentence-transformer model, and stored in memory. When you ask a question, REMI rewrites it into a standalone search query, retrieves the top 8 most relevant chunks, and passes only those to the LLM along with the conversation history.

Technology Stack

- Backend: Python + FastAPI
- PDF parsing: PyMuPDF (fitz)
- Office docs: python-docx, python-pptx
- Embeddings: sentence-transformers (all-MiniLM-L6-v2)
- Vector search: In-memory numpy cosine similarity
- LLM: OpenAI GPT API (default: gpt-5.4-nano)
- Frontend: Vanilla HTML, CSS, JavaScript (liquid-glass UI)
- Environment: python-dotenv for configuration

Architecture Notes

Everything runs in a single, readable project with a /backend folder for the FastAPI application and a /frontend folder containing one self-contained HTML file. No build step, no framework dependencies on the frontend, and no external vector database - keeping the stack minimal and easy to run locally.

Configuration

Create a .env file in the project root and add your OpenAI API key:

```
OPENAI_API_KEY=your_key_here
```

Model name and base URL are also configurable via OPENAI_MODEL and OPENAI_BASE_URL.